

Ch6 Wang's Critique: Optimal Editing & Null Distribution

Source: reading_pack/md/A06_Wang.md | Full textbook: TEXTBOOK_Stability_Identifiability.md

6 Wang

6.1 ITI

ITI $\theta = \mu - \mu$ heuristic Wang

6.2

head TruthfulQA

TruthfulQA (x, y, y) **IPO**

$$\max \sum [\log(\pi_\phi(y|x)/\pi(y|x) / \pi_\phi(y|x)/\pi(y|x)) - \tau^{1/2}]^2$$

LoRA

1. **LoRA** $r_{\text{LoRA}} = r + (W + b a) o = r_{\text{orig}} + a, o \cdot b$

• b W head

2. **Wang** $b \rightarrow W \cdot b$

...

$r_{\text{reparam}} = r_{\text{orig}} + a, o \cdot W \cdot b = r_{\text{orig}} + a, o \cdot \sum_h W_{\{h\}} \cdot b_{\{h\}}$

...

• $b_{\{h\}}$ $\theta_{\{h\}}$ a, o α

• $b_{\{h\}} = 0$ head "head"

1. Top-16 $\approx$ Full-model (Fig.2) —

2. **Random-16** $\approx$ Top-16 (Fig.3) —

3. **Head** $\approx$ Full-model **head** (Fig.4) —

4. **heuristic** Top >> Random **optimal** Top = Random

Wang p.3-4 Sec.2, p.6-9 Sec.4-6 Fig.2-4 (7)(8)

6.3

• **Fig.1** **Heuristic** Top **Info*Truth** **Random** $p = 1.6e-8$ —

• **Fig.3** **Optimal** **Random** Top —

• **Fig.4a** **24** head **5** $\approx$ Top-16
